

Response to the AI Action Plan RFI

The University of Alabama

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1 Précis

Artificial intelligence (AI) is rapidly transforming research, education, industry, and society, requiring institutions of higher education (IHE) to play a crucial role both in the advancement of AI technologies and in the preparation of future generations for an AI-driven world. This document identifies **major challenges, opportunities, and policy recommendations** that can guide the integration and development of AI in science, engineering, healthcare, cybersecurity, education, and the humanities. By fostering interdisciplinary collaboration, investing in data accessibility and AI ethics, and supporting workforce development, this report underscores the need for a strategic, responsible, and forward-thinking approach to AI research and implementation.

2 Subject Areas

2.1 AI in Science

Scope and challenges: Last year's Nobel Prizes in Physics and Chemistry have emphasized the important role of AI in recent scientific breakthroughs. The use of modern AI algorithms enables researchers and engineers to quickly analyze big data, identify complex patterns in high-dimensional datasets, predict outcomes, optimize designs, and ultimately drive innovation and groundbreaking discoveries.

Despite the exponential growth in data collection capabilities, the reality is that there is simply not enough high-quality scientific data that is ready for AI training. There is often no systematic way to report data collection methods or to document the context in which data was acquired. Sometimes, data are generated from both computational models and experimental observations, which can differ greatly in terms of format, resolution, and reliability. This discrepancy creates additional challenges when attempting to integrate diverse datasets into a unified AI framework. Another challenge is the lack of negative data, which are datasets that capture instances of failure or non-occurrence. Negative data are essential for training models that can accurately discern valid signals from noise, but are rarely

reported. In some situations, the cost of extracting, cleaning, and maintaining high-quality datasets can be very high. Many groups opt to publish only the methods by which data were generated, rather than the data themselves. Also, researchers do not receive proper credit when others utilize their data.

Opportunities: One of the major hurdles is obtaining data from diverse sources, including proprietary datasets, and ensuring that data-sharing practices benefit scientists. Novel frameworks would need to be created to ensure that researchers receive proper credit and that data contributors are rewarded, thereby encouraging more organizations to share high-quality data without compromising proprietary interests.

Another opportunity lies in developing and enforcing FAIR standards by establishing guidelines at a national level to promote a culture of transparency and quality in data. Those include finding better ways to search for good quality data in the published literature. Furthermore, data storage and management still remain a challenge as the volume of scientific data grows exponentially. Our focus also has to cover the development of innovative storage solutions to secure and manage large datasets, ensuring that the infrastructure keeps pace with the increasing demands of AI-driven research.

Policy recommendations: A critical step toward broader AI integration into science and engineering is prioritizing the conversion of research data into AI-compatible formats, and establishing mechanisms to store and manage data effectively. Specific recommendations include:

- Develop and enforce policies that promote FAIR data standards.
- Establish centralized data repositories and data science centers that serve as hubs for data storage/management and provide essential infrastructure for data curation and benchmarking, making high-quality data accessible and citable.
- Invest in academic programs and training initiatives. The University of Alabama intends to introduce an undergraduate minor (or even a full program) in data management and analytics for science.
- Support pilot projects that facilitate data sharing from proprietary sources through innovative public-private partnerships. This would involve developing frameworks that protect proprietary interests while making valuable datasets available for academic research.
- Encourage the development of advanced tools and indexing systems to search scientific literature and supplementary materials.
- Build AI models which are not as data-hungry, i.e., can generalize better with fewer data.
- Artificial intelligence can benefit from existing domain knowledge, e.g., symbolic regression, physics-informed AI, symmetry-aware AI, etc.

Beyond these policy actions, UA should aim to become a national leader in scientific data accessibility. By establishing robust data storage and management standards, UA can shape the future of AI-driven research and make high-quality, citable data available to the scientific community.

2.2 Quantum AI and Emerging Technologies

Scope and challenges: Traditional computing is approaching fundamental limitations and faces serious challenges in terms of scalability, performance, and sustainability. Quantum computing offers a promising alternative that leverages the unique properties of quantum systems to perform calculations more accurately and exponentially faster, enabling breakthroughs in fields like materials science, drug discovery, financial modeling, artificial intelligence, and cryptography. One emerging area of research is the confluence of quantum computing and machine learning: or **quantum machine learning (QML)**. Quantum machine learning focuses on quantum algorithms that perform machine learning tasks on quantum computing hardware. Despite the relatively rapid progress in the development of quantum computing and quantum machine learning, the most promising direction at the moment is offered by **hybrid** approaches in which quantum and conventional machine learning modules operate together symbiotically. However, fully quantum models are desired in the future. Open challenges include:

- Developing **practical quantum machine learning algorithms** for the existing **hybrid** computing platforms built with noisy intermediate-scale quantum (NISQ) devices.
- Optimal integration of the **quantum** and **classical** components of the hybrid QML models. What are the most promising quantum technologies for such integration?
- Identifying universally accepted benchmarks to quantify near-term **quantum AI utility**.

Opportunities: In the long term, quantum computing hardware will increasingly become a commercial commodity. Building up capabilities for quantum AI algorithm development is paramount to maintaining the USA’s leadership role in the field. The federal government can support the development of this emerging field with targeted funding opportunities and timely investments for modernizing the current workforce development and training, with an emphasis on developing a quantum AI curriculum.

Policy recommendations: The existing effort in Quantum AI algorithm development at the University of Alabama provides an opportunity to lead innovation in this area. Specific recommendations include:

- Increase the funding opportunities for complex, multidisciplinary projects on Quantum AI algorithm development and applications in targeted fields across science, engineering and medicine, whose success typically relies on teams of experts from different areas such as physics, quantum information science, mathematics, computer science, engineering and domain experts.
- Establish close relationships with the Quantum Research groups at the National Labs as well as partnerships with the industry leaders in quantum technology. Encourage incubators of AI and Quantum research and spin-off technologies.
- Fund National Research Traineeship efforts specifically in fundamental and quantum AI algorithm development. Facilitate joint opportunities that combine expertise in academia and industry in quantum AI technology.

2.3 AI for Cyberinfrastructure & Cybersecurity

Scope and challenges: The rapid expansion of AI-driven research requires robust infrastructure serving as an **AI Research Testbed** capable of supporting high-performance computing (HPC), scalable storage solutions, and secure systems.

- **Future Proofing** - AI is a very dynamic field, and computing resource needs are skyrocketing with the introduction of new generative AI models. Many new researchers are now entering the field of AI. All of this presents a challenge in forecasting and meeting short-, mid-, and long-term requirements.
- **Technical architecture** - Researchers interact with HPC systems in different ways. Hence, cyberinfrastructure must be flexible enough to accommodate diverse user preferences and levels of technical expertise.
- **Resource Allocation** - Computational needs vary across researchers and projects. Developing an optimal adaptive strategy for resource allocation remains a challenge.
- **Security and Privacy** - AI-driven systems are inherently vulnerable to security breaches, model exploitation and data privacy issues. Prioritizing and enforcing effective security countermeasures is critical to protecting infrastructure, models, and sensitive datasets.
- **Industry Partnerships** - The university must seek industry collaborations, while ensuring compliance with best practices and fostering innovation.

Opportunities: To fully leverage AI's potential in cyberinfrastructure and cybersecurity (C&C), UA must create an integrated, forward-thinking framework that supports both research and real-world applications. Key opportunities include:

- **Assistance with AI Research Needs** - UA should establish a one-stop shop that provides researchers with a full spectrum of training, documentation, hands-on workshop tutorials, and technical support for AI-driven research.
- **Interdisciplinary Collaboration** - UA should take initiative by forming interdisciplinary teams of collaborators and incentivize innovative ideas.
- **Inference API** - Large language models have become a commonplace tool for AI research, hence, the HPC at UA should consider providing inference API services.

Policy recommendations: Specific recommendations on C&C include:

- **AI-aided workflow** - Explore AI to automate existing workflows. Partnering with state agencies, local businesses and other stakeholders can help identify high-impact use cases.
- **AI in Education** - Strengthen AI literacy through partnerships with local magnet schools, community colleges and HBCUs to ensure workforce readiness.
- **Accountability of AI Models** - Promote awareness of AI accountability, transparency and fairness by carefully selecting models and integrating educational programs focused on ethical AI use.

2.4 AI in Water, Environmental, and Earth Sciences

Scope and challenges: The role of AI in Earth and Environmental Sciences (EES) has expanded significantly in the past decade. The increasing volume of AI-related EES publications reflects AI’s growing influence, particularly in areas that rely on remote sensing data. The ability to utilize large data sources has led to many unanticipated insights and discoveries throughout EES.

One of the most pressing challenges is determining the primary objectives AI should address, as many environmental goals are inherently in conflict. For example, habitat restoration and flood control often require opposing approaches. In some ecosystems, floods are a natural and necessary part of the ecological balance, yet AI-driven solutions designed to mitigate flooding may inadvertently disrupt these systems. Multi-objective optimization can help balance such competing priorities.

Another challenge is the plethora of scales on which EES phenomena occur, the complex interactions crossing scales and the data management that comes along with those interactions. This includes temporal scales ranging from seconds to centuries, and spatial scales ranging from sub-meter to entire continents. Another corresponding issue is the variety of scales at which predictions are needed for users of AI models in EES. These are challenges that make way for research opportunities and we can overcome them.

Opportunities: AI has the potential to significantly enhance forecasting accuracy and resource management in EES. In the hydrosphere, for instance, our ability to forecast floods and droughts has improved due to recent scientific advances in machine learning. This creates opportunities for enhanced hazard forecasting at the national level, attracting investment from both government and private industry.

Natural resources management has opportunities to take advantage of better prediction and optimization that comes with AI. One specific opportunity is synthesizing large-scale environmental datasets and providing actionable insights for management. Another opportunity is balancing competing resource demands, but this requires overcoming the challenge of defining primary objectives.

Policy recommendations:

- Prioritize training on **high quality data**. Given the wide range of uncertainty in EES, consideration of data sources, limitations and reliability should be made.
- Prioritize **generalization to new scenarios** beyond the range of training data. Evaluation metrics should represent applications to unknown conditions.
- Prioritize **assessments of uncertainty** and accounting for unknowns. This may include uncertainty reporting, sensitivity analysis, and integration of probabilistic approaches.
- Prioritize **operationalization** of the most promising AI methods for increasing forecasting skill and optimizing natural resources management.

2.5 AI in Transportation

Scope and challenges: The integration of AI in transportation introduces complex technical, operational, and policy considerations, particularly in ensuring trust and reliability in predictive modeling for control and operations. AI-driven systems depend on high-quality ground truth data, yet gaps in sensor coverage, data accuracy, and environmental variability introduce biases that can compromise decision-making. Furthermore, AI models must function within the constraints of physical reality, where uncertainties in input data, model assumptions, and dynamic operational conditions can lead to errors in prediction and control, potentially compromising safety and efficiency.

A critical challenge is ensuring trust, adaptability, and explainability in AI-powered transportation systems. Many AI models operate as “black boxes,” limiting the ability of engineers, policymakers, and end-users to understand or validate decisions. Liability concerns further complicate AI adoption—who is responsible when AI-driven transportation systems fail? Without clear insights into model reasoning, trust in AI-driven transportation systems remains a significant barrier to widespread implementation. Addressing these issues requires structured oversight, human-AI interaction research, and regulatory frameworks that ensure responsible deployment.

Opportunities: AI presents transformative opportunities for transportation by enabling predictive analytics, real-time traffic optimization, infrastructure resilience, and autonomous systems. Domain-specific large language models (LLMs) can integrate insights across engineering, environmental science, hydrology, and operations management, allowing for more holistic mobility solutions. This capacity to operate at the interface of multiple applications not only accelerates discovery but also expands the scope of scientific inquiry, encouraging a shift from siloed investigations to integrated analyses.

Beyond research advancements, AI also provides a critical opportunity to strengthen partnerships between academia, government, and the private sector. Transportation systems are deeply intertwined with commercial industries, from logistics and freight to mobility services and infrastructure development. Integrating human factors, psychology, and behavioral science is key to ensuring safe and effective AI-human interactions in workforce development. AI can enhance driver monitoring systems, detect fatigue or distraction, and provide adaptive feedback based on cognitive workload. Behavioral AI models can also predict how users interact with autonomous systems, mitigating automation complacency and increasing trust in AI-powered mobility. These advancements ensure that AI supports rather than replaces human decision-making, fostering a seamless transition to AI-integrated transportation.

Policy recommendations: To harness AI’s potential in transportation, policy efforts should prioritize strengthening public-private partnerships, workforce development, and structured oversight. Research-driven guidance will enhance the quality and impact of AI-driven discovery in transportation and will be critical as AI becomes increasingly embedded in the control aspects of transportation system users and infrastructure.

As such, policy efforts should prioritize:

- **Data Reliability & AI Explainability** – Improve data-sharing frameworks and promote transparent, interpretable AI models to enhance trust and accountability.

- **Human-AI Interaction Research** – Invest in cognitive science and human factors studies to optimize AI-driven systems for various users.
- **Liability & Safety Standards** – Develop clear legal frameworks defining AI accountability in autonomous vehicles and intelligent infrastructure.
- **Public-Private Partnerships** – Strengthen collaboration between academia, industry, and transportation agencies to accelerate AI-driven mobility solutions.
- **Workforce Development** – Expand AI and human factors training programs to prepare engineers, psychologists, and policymakers for AI-integrated transportation.

By integrating AI with human-centered design, interdisciplinary research, and structured oversight, we can create safer, smarter, and more resilient transportation systems while ensuring ethical and responsible AI deployment in mobility and infrastructure management. Funding should support education programs that enhance “AIQ,” equipping individuals with essential AI literacy and applied skills in data science, predictive modeling, and AI-driven decision-making. By investing in AI education, professional development, and policy oversight, policymakers can position the U.S. as a leader in AI-driven transportation innovation.

2.6 AI in Autonomous Experimentation (AE) and Robotics

Scope and challenges: AI-driven robotics significantly impacts research, industry, and society by enabling self-optimizing systems that enhance efficiency and precision. In scientific research, AE accelerates material discovery by integrating AI-driven data analysis with robotic automation, leading to rapid hypothesis testing and validation. In industry, advanced manufacturing powered by AI and robotics improves manufacturing processes, quality control, and automated supply chain management, to reduce costs and increase scalability. In society, these autonomous systems equipped with AI and robotics technologies address workforce shortages, can increase safety, and improve the quality of work environments.

Despite these advancements, several challenges hinder the integration of AI in robotics and AE. Perception and adaptability remain significant hurdles to allow robots to navigate unpredictable environments. Control and assurance mechanisms must ensure the safety and reliability of autonomy in critical applications such as autonomous vehicles, eldercare, process automation, and equipment operation. Computational constraints, including real-time processing and energy efficiency, further challenge widespread adoption. Ethical and regulatory considerations, including data privacy, liability, and human-robot teaming, must be addressed to ensure responsible deployment.

Opportunities: The integration of AI and robotics presents numerous opportunities to advance autonomy in experimentation and manufacturing. AI enhances perception, enabling robots to interpret complex sensory inputs and adapt to environmental changes. Autonomous decision-making capabilities optimize industrial processes, from damage diagnosis to adaptive motion planning. In material sciences, AE accelerates discovery cycles, significantly reducing the time required to develop new materials. Various materials systems are expected to benefit, including structural metals, functional ceramics, soft materials, and semiconductors, all of which are essential for applications in aerospace, energy storage, and biomedical

devices. Furthermore, AI-powered human-robot teaming fosters safer and more intuitive collaboration, augmenting human capabilities rather than replacing human companions.

Policy recommendations: To maximize the potential of AI in robotics and AE, policy initiatives must focus on funding, infrastructure, and workforce development. Increased investment in AI-driven robotics research through NSF, DOE, and other agencies will enhance safety, performance, and generalizability in AI-powered autonomous systems. Establishing **regional innovation hubs** will foster collaboration between academia, industry, and government and promote real-world applications of autonomous robotics and AI. Additionally, **workforce reskilling initiatives** will equip professionals with AI and robotics expertise to address labor market shifts and ensure a seamless transition to AI-augmented workplaces and laboratories. A significant emphasis must be placed on fostering **public-private partnerships** to drive commercialization and scalability of AI technologies, such that innovations transition effectively from research laboratories to industrial applications.

Strengthening U.S. competitiveness in AI and robotics is critical to ensuring domestic innovation, securing leadership in advanced manufacturing, and safeguarding national security. By investing in resilient supply chains, AI-driven automation, and cutting-edge research, the U.S. can maintain its technological edge, promote economic growth, and reinforce its position as a global leader in AI-powered robotics and manufacturing.

2.7 AI in Health and Life Sciences

Scope and challenges: Health and life sciences span from molecular biology to public health, integrating biological, technological, and environmental factors. Despite advancements, critical challenges persist. At the molecular level, DNA, proteins, and biomolecules drive cellular processes, forming tissues, organs, and systems essential for homeostasis. Public health research tracks disease patterns and informs strategies, yet disparities in healthcare access remain. Cutting-edge technologies like DNA sequencing, MRI, and wearable devices enable precise health monitoring, but gaps persist in integrating these insights into effective care. AI, bioinformatics, and biomedical engineering are revolutionizing diagnostics, drug discovery, and medical devices, while knowledge representation enhances data integration and decision-making. However, addressing social determinants of health is just as crucial as technological innovation. Bridging these gaps across biological and societal scales is imperative for advancing effective healthcare.

Integrating AI across these dimensions can foster a holistic, intelligent health ecosystem. Self-explaining and autonomous AI systems enhance personalized care, while multi-scale AI reasoning frameworks unify molecular insights, clinical data, and environmental factors for a more comprehensive understanding of health. AI-powered decision-making platforms empower clinicians, researchers, and policymakers to optimize health outcomes efficiently. Additionally, human-AI collaboration is advancing diagnostics, treatment, and disease prevention in a responsible and ethical manner. By leveraging these innovations, healthcare can become more precise, accessible, and proactive, addressing complex challenges across biological and societal scales.

To drive these advancements, federal policies must prioritize investments in AI grounding, alignment, and continuity, alongside efforts to enhance workflow transparency and ensure

privacy-preserving healthcare pipelines. Equipping healthcare professionals with AI literacy and establishing policies that align AI-driven solutions with patient-centered care are essential for fostering ethical, effective, and humane AI adoption in healthcare. Specific challenges are:

- **Grounding:** ensures AI models are biologically and physically sound across all dimensions of health and life sciences, guaranteeing reliability in clinical decision-making.
- **Alignment** prioritizes patient-centered health outcomes over cost-driven objectives and broader societal benefits, reinforcing ethical AI adoption.
- **Continuity** bridges the gap between research and policy through clinical training, insured coverage, and regulatory frameworks at both state and federal levels.

Opportunities: A unified data and knowledge fabric enables the seamless integration of molecular, clinical, and environmental health data while ensuring privacy through federated learning and secure computation. Multi-scale AI reasoning frameworks allow AI to process and infer insights across biological hierarchies, linking molecular markers to systemic health and population trends. AI-augmented decision-making platforms enhance real-time, evidence-based recommendations by leveraging predictive analytics, digital twins, and clinical decision support systems. Additionally, human-AI collaborative intelligence ensures that AI enhances rather than replaces human expertise, facilitating explainable AI-assisted diagnostics, robotic surgery, and patient engagement tools.

Policy recommendations:

- Establish AI-driven health data infrastructures to support fundamental research to enhance the interpretability of multimodal healthcare data at the individual level, ensuring AI-driven insights are robust and explainable.
- Invest in research to AI-enabled decision-making and improving AI-transparency in healthcare workflows, integrating scientific interpretability, patient-centered outcomes, and research-to-policy implementation.
- Enforce privacy-enhanced AI governance and equitable AI-driven healthcare pipelines that uphold ethical standards and ensure access to high-quality care across patient populations.
- Promote AI literacy and workforce development, equipping healthcare professionals with the skills to navigate AI-integrated healthcare systems.
- Encourage AI-driven healthcare solutions that expand access to rural populations, ensuring telehealth and predictive diagnostic models prioritize patient well-being.

2.8 AI for Humanities, Arts, and Social Sciences

Scope and challenges: Artificial intelligence is reshaping research and practice across disciplines, yet its integration into the Humanities, Arts, and Social Sciences (HASS) presents unique challenges and opportunities. HASS disciplines encompass the study of human culture, society, and behavior, engaging with complex, qualitative, and interpretive methods that often differ from the data-driven paradigms dominant in AI research. While AI offers

powerful tools for analyzing texts, artifacts, and behaviors, a digital research divide threatens to limit its transformative potential in HASS fields.

A key challenge is the digital research divide—distinct from broader digital access disparities—which affects access to AI tools, training, and computational infrastructure. Many HASS scholars lack the resources or institutional support to incorporate AI into their research, creating disparities in the ability to engage with emerging technologies. Furthermore, disciplinary barriers constrain cross-, trans-, and interdisciplinary research, making collaboration with computational fields difficult. Cultural heritage preservation also faces hurdles, as AI-driven reconstruction and analysis require high-quality datasets, expertise, and long-term funding—resources that are often scarce. Additionally, while AI can enhance efficiency, its implementation in HASS research demands substantial investment in training, new data collection methods, and human oversight to ensure ethical and interpretive integrity. Without these considerations, AI risks reinforcing rather than alleviating resource constraints in HASS.

Opportunities: Despite these challenges, AI holds promise for HASS. It can reconstruct lost artifacts, decode ancient texts, and enhance cultural preservation efforts, providing new modes of discovery. AI models can also support research on human behavior, creativity, and ethics, benefiting from HASS insights to improve interpretability and contextual awareness. Crucially, AI should not be framed as a cost-saving tool but as a research accelerator that complements human expertise rather than replacing it. AI can alleviate routine tasks, allowing HASS scholars to focus on complex analysis, theory-building, and interpretation. However, these benefits require sustained investment in faculty bandwidth, data resources, and collaborative partnerships to ensure AI integration strengthens, rather than disrupts, existing research methodologies.

Policy recommendations: To bridge the digital research divide, higher education institutions should invest in dedicated AI funding that includes medium-scale computing resources, data infrastructure, and long-term faculty support. Interdisciplinary AI labs should be established, alongside literacy training that equips HASS scholars with computational skills. Additionally, faculty must be given the time and institutional support necessary to fully engage with AI methodologies. By prioritizing these investments, institutions can position themselves at the forefront of AI-driven innovation in HASS.

2.9 AI in Ethics

Scope and challenges: As AI systems become increasingly embedded in education, research, industry, and daily life, ensuring their responsible development and use is critical. Without ethical safeguards, AI can reinforce existing inefficiencies, amplify misinformation, and produce unintended consequences that undermine trust and accountability. Higher education institutions have a unique responsibility to prepare students not only to use AI effectively but also to anticipate and mitigate its risks. This requires moving beyond abstract discussions of fairness and transparency and overemphasis on compliance to practical applications that help students, faculty, and professionals navigate complex ethical challenges involving competing principles, stakeholders, and interests. By prioritizing critical thinking, ethical reasoning, and responsible implementation, universities can equip future leaders to

make informed decisions about the role of AI in society.

AI ethics require attention at multiple levels: teaching, research, workforce development, and institutional governance. Several key challenges hinder the effective integration of ethics pedagogy in university settings:

- **Faculty Preparedness and Training Gaps** – Many faculty members lack confidence in teaching AI ethics or incorporating ethical discussions into their courses. Concerns include unfamiliarity with ethical frameworks, fear of controversy, and uncertainty about practical applications.
- **Limited Institutional Coordination** – Ethics-related AI efforts are often siloed across departments, leading to duplicated efforts and inconsistent messaging. There is a lack of centralized guidance or university-wide standards for integrating AI ethics into curricula.
- **Moving Beyond Principles to Practice** – AI ethics education is often limited to abstract discussions of fairness, transparency, and bias. Universities must develop methods for teaching applied AI ethics through case studies, project-based learning, and workforce-aligned curricula.
- **Slow Curriculum Development Processes** – Approvals for new courses and programs are slow, making it difficult to adapt to the rapidly evolving ethical landscape of AI.
- **Lack of Alignment with Broader AI Workforce Needs** – Ethical considerations in AI are not always well integrated into technical AI training or workforce preparation programs, limiting students' ability to apply ethical reasoning in industry contexts.
- **AI and the Digital Divide** – Institutions must address the ethical dimensions of AI's role in widening gaps in access to education, workforce training, and research opportunities, particularly in rural and under-resourced areas.
- **Undergraduate versus Graduate Needs** – Undergraduate students require broad AI literacy and ethics education, integrated into general education and introductory courses. Graduate students, particularly in technical fields, need more targeted interventions focused on applied ethics in research, professional practice, and policy development.

Opportunities: The University of Alabama is well-positioned to take a leading role in AI ethics education by leveraging interdisciplinary expertise, existing AI initiatives, and faculty-driven projects. Key opportunities include:

- **Developing AI Literacy and Ethics Courses** – A foundational AI literacy course with an ethics component could be implemented for undergraduates, while graduate students could engage with discipline-specific AI ethics training tailored to their research and professional trajectories.
- **Creating Applied AI Ethics Modules** – Modular, field-specific AI ethics training could be developed for STEM, humanities, business, and social sciences programs, ensuring that ethical considerations are embedded across curricula.

- **Faculty Incentives for AI Ethics Training** – Faculty need structured support to engage with AI ethics training and integrate it into their teaching. Workshops, AI “sandbox” events, and microcredentialing in AI ethics pedagogy could help bridge this gap.
- **Collaboration Across Departments and Institutions** – UA can partner with other universities and community colleges to expand AI ethics education beyond individual programs.
- **Strengthening Industry and Workforce Connections** – AI ethics should be integrated into career readiness programs, with partnerships across industry, government, and nonprofit sectors to ensure students receive real-world exposure to ethical dilemmas in AI.
- **Developing a Research Hub for AI Ethics Education** – A centralized AI ethics research initiative at UA could study best practices for teaching AI ethics, workforce applications, and the impact of AI on different learning and professional environments.
- **Building and Supporting Ethical Cultures in AI** – Beyond individual courses, universities must cultivate a broader ethical culture around AI usage, development, and implementation. This requires integrating ethics discussions into faculty training, institutional policies, and research initiatives. The Office of Research Ethics and Compliance, for example, can play a central role in supporting faculty and students in navigating ethical challenges related to AI.

Policy recommendations: To support AI ethics education and responsible AI deployment, UA recommends the following actions to NSF/OSTP:

- **Increase Funding for AI Ethics Education Research** – NSF should prioritize research funding for interdisciplinary AI ethics education, including studies on effective teaching strategies, workforce applications, and long-term impacts.
- **Support for Interdisciplinary AI Ethics Curricula** – Federal funding should enable universities to develop and scale AI ethics courses, stackable microcredentials, and training modules tailored to different fields of study.
- **Investment in Faculty Training and Professional Development** – NSF should fund AI ethics training programs for faculty, with incentives for integrating ethics discussions into technical and non-technical courses.
- **Research on AI and Access to Education and Workforce Preparation** – AI ethics research should address gaps in AI education and training, ensuring that students across different regions and institutional types can engage with AI literacy.
- **Institutional Collaboration and Infrastructure Development** – NSF should support the creation of centralized AI ethics resource hubs at universities, fostering collaboration between departments and external partners.

2.10 AI in Education and Workforce Development

Scope and challenges: Significant challenges in AI education and workforce development at UA include fostering AI literacy and skills for faculty, staff, and students while addressing

faculty hesitation due to stigma, fear, or being overwhelmed. A foundational AI course may be necessary, but curriculum development remains slow, with approval processes hindering responsiveness. Collaboration with HPC resources and workforce partnerships could enhance AI integration, but faculty need training to leverage these tools effectively. Addressing these challenges will require a coordinated, cross-campus effort to align AI education with workforce needs while ensuring ethical considerations are embedded throughout the curriculum.

Opportunities: The University of Alabama has numerous opportunities to enhance AI education and workforce development by expanding AI literacy, integrating ethics into curricula, and aligning programs with industry needs. To maximize these efforts, the university must explore innovative approaches and establish a centralized infrastructure to coordinate AI-related initiatives. Key opportunities include the following:

- Expand AI literacy courses and faculty training incentives, including AI-focused workshops, to ensure foundational knowledge and promote AI education. Integrate AI ethics into the curriculum to prepare students for responsible AI use.
- Align educational programs with workforce demands by identifying industry needs, establishing AI learning outcomes based on career clusters, and creating tailored educational pathways. Explore alternative learning models like stackable micro-credentials for flexible skill development.
- Strengthen AI workforce connections through industry, community, and state partnerships, leveraging initiatives such as Office of Teaching Innovation and Digital Education (OTIDE)/CrossingPoints to support young adults with intellectual disabilities. Expand internships and special education funding to enhance workforce readiness.
- Establish a centralized university infrastructure to track AI-related courses and resources, improving coordination and reducing redundancy. Ensure rapid curriculum updates to keep pace with AI advancements and expand workforce initiatives to support reskilling, upskilling, and K-12 AI programming across Alabama and nationally.

Policy recommendations: Based on current AI initiatives and successes at UA, the university recommends that NSF/OSTP prioritize funding to enhance AI education, ethics, and workforce development. Strategic investments are needed to support interdisciplinary research, faculty upskilling, workforce-aligned training and curriculum development. Key funding priorities include the following:

- Supporting interdisciplinary research on AI teaching, ethics and applications across undergraduate and graduate programs.
- Assessing AI's impact on institutions, communities, and workforce sectors to address concerns about the digital divide.
- Scaling faculty/staff upskilling initiatives to ensure educators can effectively integrate AI into their teaching and programming.
- Integrating AI literacy and ethics into curricula to prepare students for responsible AI use and development.

- Developing workforce-aligned training programs and experiential learning opportunities that equip students with the skills needed for jobs of the future.
- Building ethical AI cultures within institutions by moving beyond compliance to applied, actionable ethics in AI design, deployment, and usage.
- Creating sustainable, equitable AI education models that keep pace with technological advancements and industry demands.
- Creating opportunities for faculty-industry connections that may lead to use-inspired research collaborations and potential commercialization.

By prioritizing these initiatives, NSF/OSTP can help institutions like UA lead in AI education and workforce development while addressing critical ethical challenges.

3 Summary

The previous discussion highlighted AI’s transformative potential across multiple domains, while also underscoring key challenges that must be addressed to ensure responsible and effective implementation. From fundamental research to applied solutions, AI is reshaping science, engineering, healthcare, and the humanities in profound ways. To harness these advancements, institutions must prioritize interdisciplinary collaboration, robust infrastructure, ethical considerations, and workforce development. Additionally, stronger **public-private partnerships (PPPs)** will be essential for accelerating AI research, technology transfer, and workforce training. The following key themes emerged as critical areas for future investment and policy focus:

- *Data Accessibility and Quality* – Many fields struggle with data-related challenges, including limited availability, inconsistent formatting, and proprietary restrictions that hinder AI’s effectiveness. Addressing these barriers requires public data repositories, standardized metadata, and the development of AI models capable of working with diverse and incomplete datasets.
- *AI and emerging quantum technologies* – The next frontier of AI is Quantum Machine Learning, which leverages the advantages of quantum computing and modern GPU architectures. QML models can address future challenges in machine learning through their high expressiveness, faster convergence, and potential for higher accuracy with a smaller number of model parameters.
- *Interdisciplinary Collaboration* – The complexity of AI applications necessitates collaboration across disciplines. Whether in quantum AI, environmental sciences, or transportation, integrating domain knowledge with AI methodologies is essential for meaningful advancements.
- *Ethics, Transparency, and Accountability* – Concerns over AI’s “black box” nature and its ethical implications were raised across multiple disciplines. Ensuring AI models are interpretable, unbiased, and aligned with societal needs will require stronger regulations, workforce training, and educational initiatives in AI ethics.

- *Workforce Development and AI Literacy* – AI education must be expanded beyond technical fields to include healthcare, social sciences, and the humanities. Faculty training, curriculum updates, and interdisciplinary programming/courses are critical to preparing students for AI-integrated careers. UA’s AI workforce efforts should aim to improve educational outcomes both internally and externally as it aligns with the state’s economic development plan and fosters industry-academia-community partnerships.
- *Cybersecurity and Infrastructure* – As AI adoption grows, so do challenges related to security, privacy, and computational resources. Institutions must invest in high-performance computing infrastructure while ensuring robust security measures for AI-driven systems. The integration of Quantum AI technologies into the existing HPC infrastructure will be a key challenge in the near-term future.
- *AI for Societal Impact* – From healthcare and environmental sciences to transportation and humanities, AI presents significant opportunities to improve quality of life, sustainability, and decision-making. However, responsible deployment and governance are key to maximizing benefits while mitigating risks.
- *AI-Driven Data Innovation* – Beyond addressing existing data limitations, AI itself presents opportunities to transform data management and utilization. Emerging AI techniques can enhance data collection, automate curation, and extract meaningful insights from unstructured or multimodal datasets. Establishing FAIR (Findable, Accessible, Interoperable, Reusable) data principles and interdisciplinary data-sharing frameworks will be critical to unlocking AI’s full potential in research and industry.
- *Autonomous Experimentation and Robotics* – AI-driven robotics has the potential to revolutionize scientific discovery, manufacturing, and industrial automation. However, challenges related to perception, adaptability, and ethical considerations must be addressed to ensure responsible and scalable deployment. Increased investment in autonomous experimentation could accelerate breakthroughs in materials science, engineering, and other domains.
- *Public-Private Partnerships (PPPs) as AI Accelerators* – Collaboration between academia, industry, and government is crucial for AI innovation. PPPs can accelerate technology transfer, ensure AI research aligns with real-world applications, and create workforce training programs that prepare students for AI-driven industries. Structured partnerships will be essential for expanding AI’s impact in sectors such as healthcare, transportation, cybersecurity, and manufacturing.

By addressing these challenges and leveraging AI’s transformative potential, institutions like the University of Alabama can drive the next generation of AI innovation. **Strategic investment in research, infrastructure, and public-private collaboration will enable universities to serve as catalysts for AI-driven advancements that push technological frontiers while ensuring societal benefits.** By fostering ethical AI development, strengthening interdisciplinary partnerships, and expanding access to AI education, academic institutions can position themselves as **national and global leaders in AI research and application, shaping a future where AI serves as a tool for scientific discovery, economic growth, and societal well-being.**